Felipe Nunes

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Embedded Systems Engineer

Education

University of British Columbia — Sep 2020 – May 2026 (Expected)
Bachelor of Applied Science in Integrated Engineering

Experience

Sarcomere Dynamics — Sep 2025 – Present
Electrical and Firmware Engineering Intern — *Vancouver, B.C.*

- Redesigned 6-layer PCB with controlled impedance routing for USB 2.0, CAN bus, and RS-485 interfaces
- Conducted hardware-in-the-loop testing to validate system performance across 10+ test scenarios

Miru Smart Technologies — May 2024 – Aug 2024
Systems Integration Intern — *Vancouver, B.C.*

- Reduced PCB footprint by 60% (6-layer, BGA SoM) and added a tuned 2.4 GHz inverted-F antenna
- Wrote C modules in Zephyr to drive electrochromic windows controller, increasing transitioning rate by 50%
- Implemented low-power sleep modes and power management for battery-operated IoT device
- Optimized firmware for ARM Cortex-M4 with real-time constraints and interrupt handling

Sarcomere Dynamics — May 2023 – Apr 2024
Electrical and Firmware Engineering Intern — *Vancouver, B.C.*

- Implemented embedded C firmware with a finite state machine to control actuators in a multi-DOF robotic hand
- Integrated multiple sensor inputs (IMU, force sensors) for real-time haptic feedback control
- Optimized control algorithms for sub-millisecond response times in safety-critical applications
- Developed custom communication protocol for synchronized multi-actuator coordination

Projects

Robotic Chessboard Code | Report — Sep 2024 – Apr 2025

- Won the Best Engineering Project award out of all final-year Capstone projects
- Designed custom PCBs and cable assemblies for a multiplexed Hall effect sensor grid to detect chess piece positions
- Implemented real-time firmware for STM32 microcontroller with interrupt-driven sensor polling
- Developed hardware abstraction layers for stepper motor control and sensor interfacing

Mountable Heads Up Display Code | Report — Oct 2022 – Apr 2023

- Engineered an ESP32-based control unit to interface with UBC Formula Electric's CAN bus
- Displayed real-time telemetry (speed, temp, lap time) on an I2C OLED using FreeRTOS task scheduling
- Designed a compact custom PCB integrated into a 3D-printed helmet-mounted enclosure

Automated Drink Dispenser Code | Report — Nov 2021 – Apr 2022

- Constructed a custom beverage dispenser using CNC-milled aluminum and a 3D printed nozzle
- Controlled dosing system with solenoid valves and load cell feedback for precision pouring
- Wrote firmware for automated calibration, stepper control, drink selection and dispensing, and LCD user interface

Leadership and Teams

Open Robotics — Sep 2024 – Present
Co-Captain (Sep 2025 - Present) and Team Lead — *University of British Columbia*

- Expanded responsibilities to include strategic planning, member recruitment, and competition preparation
- Managed cross-disciplinary team of eight engineers across mechanical, electrical, and software domains

Supermileage — Sep 2022 – Aug 2023
Electrical Team Member — *University of British Columbia*

- Redesigned and assembled a custom PCB leveraging analog sensors to detect faults in a hydrogen electric FSAE car
- Competed and won 2023 SEMA competition in the hydrogen fuel cell category at the Indianapolis Motor Speedway

Skills

Programming Languages: C/C++, Python, JavaScript, TypeScript, Assembly, VHDL, MATLAB, SQL, Shell Scripting
Operating Systems: Linux (Arch, Ubuntu, Raspbian), Windows, macOS, FreeRTOS, Zephyr
Hardware Design: Circuit design/analysis, PCB layout (Altium, KiCad), Mixed-signal design, Power management
Development Tools: GDB, OpenOCD, JTAG, Logic Analyzers, Oscilloscopes, Multimeters

Languages

French (DELF B2), Portuguese (Bilingual Proficiency), Spanish (Working Proficiency)